GlucoTwin AI

A Time-Series Digital Twin & 30-Minute Glucose Forecaster Proof of Concept

1. Executive Overview & Problem Statement

Conventional healthcare often reviews glucose readings as isolated, episodic snapshots. GlucoTwin AI demonstrates an adaptive virtual patient digital twin that combines historical context and continuous sensor time-series to update internal state and predict blood glucose 30 minutes ahead.

2. Healthcare & Digital Twin Use Case

A simulated patient continuously streams biometrics (interstitial glucose, meal carbohydrate estimates, physical exertion, sleep duration, and heart rate). The engine computes forward physiological trajectories, updates virtual twin states, and flags values outside demonstration target thresholds (70–180 mg/dL).

3. Machine Learning Baseline & Verified Evaluation Benchmarks

Model: RandomForestRegressor (120 trees, max depth 12) trained on lag features (5-min, 15-min), 30-min rolling means, and circadian temporal signals. Evaluated with a strict 80/20 chronological split to avoid time-series lookahead leakage.

Metric	Value	Evaluation Details
Mean Absolute Error (MAE)	**6.329 mg/dL**	Evaluated on 698 chronological test observations
Root Mean Squared Error (RMSE)	**7.814 mg/dL**	Demonstrates tight error bounds on synthetic trajectory
Dominant Predictive Drivers	Glucose 71.8% | Meal Carbs 6.6%	Correlates physiological carbohydrate spikes & recovery

4. Full-Stack Implementation

- **Backend (FastAPI):** RESTful endpoints for health status, twin state (/twin/demo-patient), 30-min forecast (/forecast/demo-patient), time-series history, dynamic simulation steps, and metrics.
- **Frontend (HTML5/Vanilla CSS/JS):** Full-screen edge-to-edge 3D dashboard featuring real-time SVG CGM telemetry, 3D bio-particle hologram canvas, tactile What-If simulation studio, and light/dark theme modes.

5. Open Source, Ethics & Public Accessibility

- **Public Repository:** https://github.com/tharundev-123/GlucoTwin_AI | License: MIT
- **Responsible-Use Notice:** Operates on synthetic educational data. It is not an FDA-cleared medical device and must not guide clinical diagnosis or therapy.